

Lecture - 12

$$\theta_j := \theta_j \left(1 - \alpha \frac{\lambda}{n}\right) - \frac{\alpha}{n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} \quad - (1)$$

$$\alpha > 0, \lambda > 0, n > 0$$

$$\rightarrow c = 1 - \alpha \frac{\lambda}{n} < 1$$

$$c \approx 0.9$$

Hence, we are shrinking θ_j .

Normal
Eqn...

with
Regularization

$$\hat{\theta} = (X^T X + \lambda I)^{-1} X^T y \quad - (2)$$

$$\hat{\theta} = (X^T X)^{-1} X^T y \quad - (3)$$

If $X^T X$ is Singular matrix, we can not use

Eq. (3).

$$\boxed{(X^T X)^{-1} X^T} \leftarrow X^+$$

Singular Value Decomposition

$$X = U \Sigma V^T$$

$$X = U \begin{bmatrix} \sigma_1 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & \dots & 0 \\ 0 & 0 & \sigma_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_n \end{bmatrix} V^T$$

U = Left Singular matrix

V = Right " "

$\Sigma = [\sigma_i]$, where σ_i = Singular Values

Suppose $\Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow x^T x$ has a 0 eigen value.

$\Rightarrow$ Non-invertible.

$$\Sigma^+ = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x^T = V \Sigma^+ U^T$$

$$\hat{\theta} = x^T y$$

$x \rightarrow x^T x \rightarrow \text{Eigenvalues } (\lambda_i) \rightarrow \sigma_i = \sqrt{\lambda_i}$

$$\Sigma = \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r \end{bmatrix}$$

Eigenvectors $\rightarrow v_i$

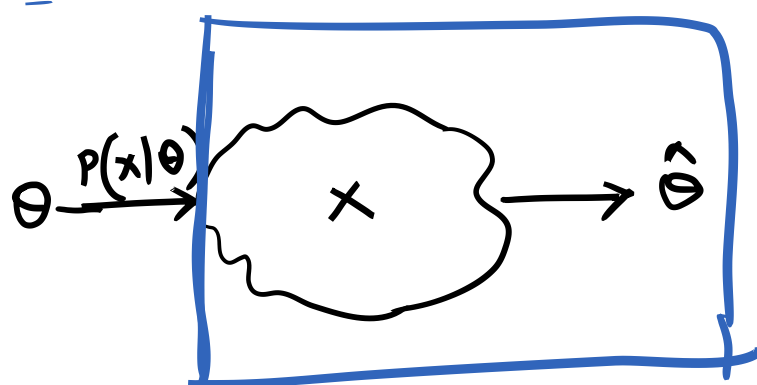
$$V = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix}$$

$x \rightarrow x x^T \rightarrow \text{Eigenvalues / Eigenvectors } U = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix}$

$$\bar{z}^T : \sigma_i \leftarrow \frac{1}{\sigma_i}$$

$$0 \leftarrow 0$$

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$



$P(x; \theta)$
Probability of x
parameterized by θ .

$P(x|\theta)$
Probability of x
Conditioned on θ

$$X = Y$$

$$\vec{y} = [y_1, y_2, \dots, y_n]^T \leftarrow P(y|\theta)$$

$$y_n \sim P(y|\theta) \quad \forall n.$$

↓
Random Variable

⇒ θ : parameter.

⇒ observations / Data samples are i.i.d

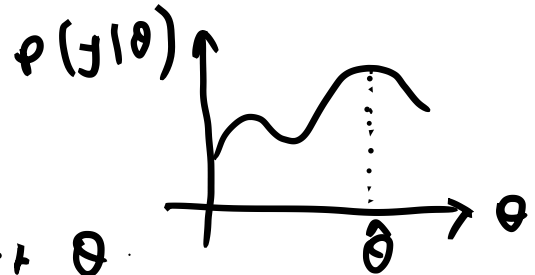
Objective is to learn parameter θ from y .

$$p(y|\theta)$$

$$= p(y_1, y_2, \dots, y_n | \theta)$$

$$= \prod_{i=1}^n p(y_i | \theta)$$

How do we estimate the best model parameter θ ?



maximize $p(y|\theta)$ w.r.t θ

$$\underbrace{p(\theta|y)}_{\text{Posterior}} = \frac{\underbrace{p(y|\theta)}_{\text{Likelihood}} \underbrace{p(\theta)}_{\text{prior}}}{\underbrace{p(y)}}_{\text{Evidence}}$$